Optimization and Curve Sketching

ANSWER KEY

Critical points and the first/second derivative tests

1. Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.

   $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, critical points $x = 1, 3$. $f''(x) = 6x - 12$: $f''(1) = -6 < 0$ (local max); $f''(3) = 6 > 0$ (local min).

2. For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

   $f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ (concave up) on $(-\infty, 0) \cup (2, \infty)$; $f'' < 0$ (concave down) on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$.

Optimization applications

3. A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.

   Let x = width (two sides), y = length (one side): $2x + y = 200$, so $y = 200 - 2x$. Area $A = xy = x(200 - 2x) = 200x - 2x^2$. $A' = 200 - 4x = 0$ gives $x = 50$, $y = 100$. Maximum area 5000 m^2 at width 50 m, length 100 m.

4. An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.

   $V(x) = x(12 - 2x)^2$, for $0 < x < 6$. $V'(x) = (12 - 2x)^2 + x \cdot 2(12 - 2x)(-2) = (12 - 2x)(12 - 6x)$. Setting $V' = 0$: $x = 6$ (rejected, gives $V = 0$) or $x = 2$. Maximum volume at $x = 2$ cm.